

# Contents

|                                             |   |
|---------------------------------------------|---|
| Introduction                                | 1 |
| Syntax and Semantics of Propositional Logic | 1 |
| Formalisation in Propositional Logic        | 2 |
| The Syntax of Predicate Logic               | 2 |
| Semantics of Predicate Logic                | 3 |
| Natural Deduction                           | 4 |

## Introduction

The following notes were compiled to refresh my logic knowledge, based on The Logic Manual (Halbach, V., 2010)

## Syntax and Semantics of Propositional Logic

The formula  $\psi$  is *logically true* iff  $\emptyset \models \psi$ , i.e. iff  $\models \psi$ . In propositional logic, a logically true formula is called a *tautology*.

**Theorem 2.14 (Semantic entailment, Logical entailment, Logically implies, p. 43).** For  $n \geq 1$ ,

$$\{\psi_i\}_{i=1}^n \models \phi \iff \left( \bigwedge_{i=1}^n \psi_i \right) \rightarrow \phi \text{ is a tautology.}$$

or in other word:

$$\{\psi_i\}_{i=1}^n \models \phi \iff \models \left( \left( \bigwedge_{i=1}^n \psi_i \right) \rightarrow \phi \right).$$

**Example.** We can prove that a propositional formula is a tautology by enumerating every possible assignment of truth values to

$$((P \rightarrow \neg Q) \wedge Q) \rightarrow \neg P.$$

Its truth table is:

| $P$ | $Q$ | $P \rightarrow \neg Q$ | $(P \rightarrow \neg Q) \wedge Q$ | $\neg P$ | $((P \rightarrow \neg Q) \wedge Q) \rightarrow \neg P$ |
|-----|-----|------------------------|-----------------------------------|----------|--------------------------------------------------------|
| T   | T   | F                      | F                                 | F        | T                                                      |
| T   | F   | T                      | F                                 | F        | T                                                      |
| F   | T   | T                      | T                                 | T        | T                                                      |
| F   | F   | T                      | F                                 | T        | T                                                      |

Since the final column contains only T, the formula is a tautology. Hence, by Theorem 2.14,

$$P \rightarrow \neg Q, Q \models \neg P.$$

## Formalisation in Propositional Logic

$L_1$  is the formal language of propositional logic: sentence letters such as  $P$ ,  $Q$ , and  $R$ ; connectives such as  $\neg$ ,  $\wedge$ ,  $\vee$ ,  $\rightarrow$ , and  $\leftrightarrow$ ; and formation rules determining which strings are sentences. For example,  $\neg P$  and  $(P \wedge Q)$  are sentences of  $L_1$ .

---

A *propositional formula* is an expression built from sentence letters and connectives according to the formation rules. A *sentence* is a formula with no free variables. A sentence may be assigned a truth value relative to an interpretation or structure, e.g. a structure  $A$  might assign  $A(P) = T$ . (p. 87)

---

In propositional logic there are no free individual variables, so every well-formed propositional formula can be treated as a sentence.  $P$ ,  $\neg P$ , and  $P \wedge Q$  are therefore called sentences of  $L_1$ , while  $\phi$  and  $\psi$  are metavariables ranging over sentences of  $L_1$ , e.g.  $\phi$  may stand for  $P \wedge Q$ .

Connectives often include  $\wedge$ ,  $\vee$ , and  $\neg$ , but a sentence-forming operator such as “it is necessarily the case that...” can also be a connective, although it is not part of classical propositional logic and is not truth-functional (see below)

[ADD THE ZENO EXAMPLE BELOW HERE]

---

A *direct subsentence* is the sentence “at the next level down to” the sentence in question, e.g. the direct subsentence of  $\neg(P \wedge Q)$  is  $P \wedge Q$ , whereas the direct subsentences of  $P \wedge Q$  are  $P$  and  $Q$ .

**Characterisation 3.1 (Truth Functionality, p.54)** A connective is *truth-functional* iff the truth-value of the compound sentence cannot be changed by replacing a direct subsentence with another sentence having the same truth-value.

The usual connectives  $\wedge$ ,  $\vee$ ,  $\neg$ , etc are always truth functional.

**Example (Not truth functional).** Let  $P$  be the sentence “ $2 + 2 = 4$ ,” and let  $Q$  be the sentence “London is the capital of the UK.” Both  $P$  and  $Q$  are true. However, under the usual interpretation of  $\Box$  as “it is necessarily the case that,”  $\Box P$  is true, whereas  $\Box Q$  is false: London is the capital, but historically it could have turned out otherwise. Thus, replacing  $P$  with the equally true sentence  $Q$  changes the truth value of the compound sentence: although  $Q$  is true,  $\Box Q$  is false because London is not *necessarily* the capital of the UK. Therefore,  $\Box$  is not truth-functional.

---

**Basics Recap.** Recall  $A \rightarrow B$  means “if  $A$  is true, then  $B$  follows as a consequence”. So,  $A$  is sufficient for  $B$  because  $A$  is enough to obtain  $B$  as a consequence. Whereas  $B$  is necessary for  $A$  because  $B$  follows from  $A$ .

---

Scope ambiguities can arise so we must use parentheses appropriately, e.g.  $P \wedge (Q \vee R)$  versus  $(P \wedge Q) \vee R$ .

## The Syntax of Predicate Logic

Consider

$P$ : Zeno is a tortoise.  $Q$ : All tortoises are toothless. Therefore  $R$ : Zeno is toothless.

This argument is not propositionally valid since the two premisses and conclusion are formalised as three sentence letters  $P$ ,  $Q$  and  $R$  respectively, but  $P, Q \not\models R$ .

*Propositional logic treats each complete sentence as an atomic unit*, i.e. there is no propositional connective linking the three sentence letters, and propositional logic does not look inside them to see the recurrence of Zeno, tortoise, and toothless.

---

We can represent formulae using e.g.  $Px$  or  $Pxy$ , sometimes we include the *arity-indices*, e.g.  $P^1x$  or  $P^2xy$  to indicate the number of variables accepted by  $P$ .

---

**Example** (p.89) Converting English sentences to predicate logic: “All frogs are amphibians” translates as “for all  $x$  (if  $x$  is a frog, then  $x$  is an amphibian)” or in symbols  $\forall x(P(x) \rightarrow Q(x))$ , with the predicate letters (not sentences)  $P$ : ... is a frog, and  $Q$ : ... is an amphibian.  $P$  and  $Q$  become formulae when supplied with a variable, e.g.  $P(x)$ ,  $Q(x)$ . When a variable is bound, i.e.  $\forall xP(x)$ ,  $\exists xP(x)$ , the formula becomes a sentence.

**Example** (p.92) “If it’s raining then Bill reads a book or a newspaper” translates as  $P \rightarrow \exists x(P^2ax \wedge (Qx \vee Rx))$  with  $P$ : it’s raining,  $Q^1$ : ... is a book,  $R^1$ : ... is a newspaper, and  $P^2$ : ... reads ... .

---

## Semantics of Predicate Logic

Meaning of “logically true”:

A sentence of the language  $L_2$  is said to be *logically true*  $\iff$  it is true under any interpretation of the constants and predicate letters.

Meaning of “valid”:

An argument in  $L_2$  will be defined to be valid  $\iff$  there is no interpretation under which the premisses are all true and the conclusion is false, i.e.  $\Gamma \models \phi$  where  $\Gamma$  is the set of premisses and  $\phi$  the conclusion.

Meaning of “structure”:

A *structure*  $\mathcal{A}$  consists of a non-empty *domain*  $D$  and an interpretation function  $I$ . The domain  $D$  contains the objects under discussion, while  $I$  assigns:

- each constant  $a$  an element  $I(a) \in D$ ;
- each unary predicate letter  $P$  a subset  $I(P) \subseteq D$ ;
- each  $n$ -ary predicate letter  $R$  an  $n$ -ary relation  $I(R) \subseteq D^n$ .

For example, let  $D = \mathbb{Z}$ , and suppose  $a$  means zero,  $P$  means “... is even,” and  $R$  means “... is less than ...”. Then

$$\begin{aligned} I(a) &= 0, \\ I(P) &= \{n \in \mathbb{Z} : \exists k \in \mathbb{Z} (n = 2k)\}, \\ I(R) &= \{(m, n) \in \mathbb{Z}^2 : m < n\}. \end{aligned}$$

Consequently,  $P(a)$  is true because  $I(a) \in I(P)$ . If  $I(b) = 3$ , then  $R(a, b)$  is true because  $(0, 3) \in I(R)$ .

---

**Semantic Value** (see p. 99). Let  $|e|_{\mathcal{A}}^{\alpha}$  be the *semantic value* of the expression  $e$  in the  $\mathcal{L}_2$ -structure  $\mathcal{A}$  under the variable assignment  $\alpha$  (i.e.  $\alpha$  is a map over  $\mathcal{A}$ ). Let  $\mathcal{A} = (D_{\mathcal{A}}, I_{\mathcal{A}})$ , where  $D_{\mathcal{A}}$  is the domain and  $I_{\mathcal{A}}$  is the interpretation function that assigns semantic values to the non-logical symbols (i.e. constants and predicates):

- **Constant**  $t$ :  $|t|_{\mathcal{A}}^{\alpha} \in D_{\mathcal{A}}$  assigned to  $t$  by  $\mathcal{A}$ , e.g.  $|c|_{\mathcal{A}}^{\alpha} = 5$  if  $c = 5$  by  $\mathcal{A}$ .
- **Variable**  $v$ :  $|v|_{\mathcal{A}}^{\alpha} \in D_{\mathcal{A}}$  assigned to  $v$  by  $\alpha$ , e.g.  $|x|_{\mathcal{A}}^{\alpha} = 7$  if  $\alpha(x) = 7$ .
- **Sentence letter**  $P$ :  $|P|_{\mathcal{A}}^{\alpha} \in \{T, F\}$  assigned directly by  $\mathcal{A}$ ;  $\alpha$  has no effect.
- **Formula / Sentence**  $\phi$ :  $|\phi|_{\mathcal{A}}^{\alpha} \in \{T, F\}$ ;  $\alpha$  matters if  $\phi$  has free variables.  $|\phi|_{\mathcal{A}}^{\alpha} = T$  means “ $\alpha$  satisfies  $\phi$  in  $\mathcal{A}$ ”, sometimes written  $\mathcal{A} \models \phi[\alpha]$ .  $\phi$  is a sentence if it has no free-variables.
- **Unary predicate**  $\phi$ :  $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}$ , e.g. if  $E$  = “is even”, then  $|E|_{\mathcal{A}}^{\alpha} = \{\dots, -2, 0, 2, 4, \dots\}$ .
- **Binary predicate**  $\phi$ :  $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}^2$ , e.g. if  $L$  = “<”, then  $(2, 5) \in |L|_{\mathcal{A}}^{\alpha}$ .

- **3-ary predicate**  $\phi$ :  $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}^3$ , e.g. if  $S = "x + y = z"$ , then  $(2, 3, 5) \in |S|_{\mathcal{A}}^{\alpha}$ .
- **$n$ -ary predicate**  $\phi$ :  $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}^n$ , i.e. a set of ordered  $n$ -tuples, e.g. if  $\phi(x_1, \dots, x_n) = "x_1 + \dots + x_n = a"$ , then  $(x_i)_{i=1}^n \in |\phi|_{\mathcal{A}}^{\alpha} \iff \sum_{i=1}^n x_i = a$ .

**Definition 5.2 (Satisfaction, pp.100-104):**

- **Atomic (no connectives or quantifiers) formula**:  $|\Phi t_1 \dots t_n|_{\mathcal{A}}^{\alpha} = T$  iff  $(|t_1|_{\mathcal{A}}^{\alpha}, \dots, |t_n|_{\mathcal{A}}^{\alpha}) \in |\Phi|_{\mathcal{A}}^{\alpha}$ , where  $\Phi$  is an  $n$ -ary predicate letter, e.g.  $Px_1x_2\dots x_n$ , with  $n \geq 1$ , and each  $t_i$  is either a variable or constant.
- **Negation**:  $|\neg\phi|_{\mathcal{A}}^{\alpha} = T$  iff  $|\phi|_{\mathcal{A}}^{\alpha} = F$ .
- **Conjunction**:  $|\phi \wedge \psi|_{\mathcal{A}}^{\alpha} = T$  iff  $|\phi|_{\mathcal{A}}^{\alpha} = T$  and  $|\psi|_{\mathcal{A}}^{\alpha} = T$ .
- **Disjunction**:  $|\phi \vee \psi|_{\mathcal{A}}^{\alpha} = T$  iff  $|\phi|_{\mathcal{A}}^{\alpha} = T$  or  $|\psi|_{\mathcal{A}}^{\alpha} = T$ .
- **Conditional**:  $|\phi \rightarrow \psi|_{\mathcal{A}}^{\alpha} = T$  iff  $|\phi|_{\mathcal{A}}^{\alpha} = F$  or  $|\psi|_{\mathcal{A}}^{\alpha} = T$ , the reason for which can be seen directly from the truth table for  $\rightarrow$ :  $TTT$ ,  $TFF$ ,  $FTT$ ,  $FFT$ .
- **Biconditional**:  $|\phi \leftrightarrow \psi|_{\mathcal{A}}^{\alpha} = T$  iff  $|\phi|_{\mathcal{A}}^{\alpha} = |\psi|_{\mathcal{A}}^{\alpha}$ .
- **Existential quantifier**:  $|\exists v\phi|_{\mathcal{A}}^{\alpha} = T$  iff there is some value in  $D_{\mathcal{A}}$  that can be assigned to  $v$ , while leaving the values of all other variables fixed by  $\alpha$ , for which  $\phi$  is true. Formally,  $|\exists v\phi|_{\mathcal{A}}^{\alpha} = T \iff |\phi|_{\mathcal{A}}^{\beta} = T$  for some assignment  $\beta$  such that  $\beta(u) = \alpha(u)$  for every variable  $u \neq v$ .
- **Universal quantifier**:  $|\forall v\phi|_{\mathcal{A}}^{\alpha} = T$  iff  $\phi$  is true for every possible value in  $D_{\mathcal{A}}$  assigned to  $v$ , while leaving the values of all other variables fixed by  $\alpha$ . Formally,  $|\forall v\phi|_{\mathcal{A}}^{\alpha} = T \iff |\phi|_{\mathcal{A}}^{\beta} = T$  for every assignment  $\beta$  such that  $\beta(u) = \alpha(u)$  for every variable  $u \neq v$ .

[TODO: **Examples**: show the examples i read through on pp.105-108, i understood them all.]

**Definition 5.7 (Logical truth, contradiction, equivalence & semantic consistency, p.108):**

- **Logically true**: A sentence  $\phi$  is logically true iff  $\phi$  is true in every  $\mathcal{L}_2$ -structure  $\mathcal{A}$ ; i.e.  $\mathcal{A} \models \phi$  for every  $\mathcal{A}$ .
- **Contradiction**: A sentence  $\phi$  is a contradiction iff  $\phi$  is false in every  $\mathcal{L}_2$ -structure  $\mathcal{A}$ ; i.e.  $\mathcal{A} \not\models \phi$  for every  $\mathcal{A}$ .
- **Logically equivalent**: Sentences  $\phi$  and  $\psi$  are logically equivalent iff they are true in exactly the same  $\mathcal{L}_2$ -structures; i.e.  $\mathcal{A} \models \phi \iff \mathcal{A} \models \psi$  for every  $\mathcal{A}$ .
- **Semantically consistent**: A set  $\Gamma$  of  $\mathcal{L}_2$ -sentences is *semantically consistent* iff there is at least one  $\mathcal{L}_2$ -structure  $\mathcal{A}$  in which every sentence in  $\Gamma$  is true; i.e. there exists some  $\mathcal{A}$  such that  $\mathcal{A} \models \phi$  for every  $\phi \in \Gamma$ .  $\Gamma$  is *semantically inconsistent* iff no such  $\mathcal{A}$  exists.

**Definition 5.8 (Validity; Valid Argument, p.109):** Let  $\Gamma$  be a set of  $\mathcal{L}_2$ -sentences (premises) and  $\phi$  an  $\mathcal{L}_2$ -sentence (conclusion). The argument is *valid* iff there is no  $\mathcal{L}_2$ -structure  $\mathcal{A}$  in which every sentence in  $\Gamma$  is true and  $\phi$  is false. Equivalently, every  $\mathcal{L}_2$ -structure that makes all the premises true also makes the conclusion true. This is written  $\Gamma \models \phi$  and read “ $\Gamma$  logically implies  $\phi$ ” or “ $\phi$  logically follows from  $\Gamma$ .” We also say “semantically entails” or “logically entails” for “logically implies”. **In other words:** There is no structure  $\mathcal{A}$  in which all the premises in  $\Gamma$  are true but the conclusion  $\phi$  is false.

**Note:** This is closely related to *tautology*, but that term is usually reserved for propositional logic. In predicate logic we instead say a sentence is *logically true* iff it is true in every structure.

An  $\mathcal{L}_2$  structure  $\mathcal{A}$  is a *counterexample* to an argument iff all premises of the argument are true in  $\mathcal{A}$  and the conclusion is false in  $\mathcal{A}$ .

[TODO: work through the counterexample examples]

## Natural Deduction

**Note:** the following notes use the **bussproof** which works with Obsidian, but this is easily adapted to ebproof or plain LaTeX instead.

**Basics.**

- As example,  $P \wedge Q, P \rightarrow R \vdash P \vee Q$  reads as: the premises (left) *prove/derive* ( $\vdash$ ) the conclusion  $P \vee Q$ , which is different from *logical implication* ( $\models$ ) - see below.
- The premises  $P \wedge Q$  and  $P \rightarrow R$  are free to use throughout the deduction.
- As we proceed we may introduce temporary assumptions, e.g.  $P$ , but when an assumption is discharged by a rule we enclose it in square brackets  $[P]$  to show it has been discharged.
- Various rules are applied to deduce conclusions from premises, and so the proof tree builds until we reach the desired final conclusion. The rules are:
- $\wedge$  Intro (*introduction*),  $\wedge$  Elim (*elimination*)
- $\vee$  Intro<sub>1</sub> (*left*),  $\vee$  Intro<sub>2</sub> (*right*),  $\vee$  Elim
- $\rightarrow$  Intro,  $\rightarrow$  Elim
- $\neg$  Intro,  $\neg$  Elim

**Theorem (Soundness).**  $\Gamma \vdash \varphi \Rightarrow \Gamma \models \varphi$ .

If  $\varphi$  is provable from  $\Gamma$ , then  $\varphi$  is logically implied by  $\Gamma$ .

**Theorem (Completeness).**  $\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi$ .

If  $\varphi$  is logically implied by  $\Gamma$ , then  $\varphi$  is provable from  $\Gamma$ .

**Theorem (Adequacy).**  $\Gamma \models \varphi \iff \Gamma \vdash \varphi$ .

$\Gamma \vdash \varphi$  means: there is a formal proof of  $\varphi$  from  $\Gamma$  using the allowed rules of deduction. Whereas  $\Gamma \models \varphi$  means: in every interpretation where all sentences in  $\Gamma$  are true,  $\varphi$  is also true.

For example  $P, P \rightarrow Q \vdash Q$  because  $Q$  can be formally derived by modus ponens, and  $P, P \rightarrow Q \models Q$  because there is no interpretation in which both premises are true and  $Q$  is false.

So  $\vdash$  is about proofs, whereas  $\models$  is about truth in all models, i.e. logical consequence; when there are no premises,  $\emptyset \models \varphi$ , or simply  $\models \varphi$ , expresses logical validity, analogous to a *tautology* in propositional logic.

**Example 6.3**  $P \vee Q, P \rightarrow R \vdash R \vee Q$ . **Proof.**

$$\frac{\frac{P \vee Q}{\frac{[P] \quad P \rightarrow R}{R} (\rightarrow \text{Elim})} (\vee \text{Intro}_1) \quad \frac{[Q] \quad P \rightarrow R}{R \vee Q} (\vee \text{Intro}_2)}{R \vee Q} (\vee \text{Elim})$$

**Explanation.** The premises are  $P \vee Q$  and  $P \rightarrow R$ . Since we know  $P \vee Q$  we can assume  $P$  in one branch and  $Q$  in another and see if we can derive the conclusion  $R \vee Q$  from each. With  $P$  we know  $R$  since  $P \rightarrow R$ . Now we can trivially introduce  $R \vee Q$ . On the other hand, if we assume  $Q$  then we can trivially deduce  $R \vee Q$ . Finally, since  $P \vee Q$ , we have covered both branches and therefore  $R \vee Q$ .

**Example 6.4**  $\neg(P \rightarrow Q) \vdash \neg Q$ . **Proof.**

$$\frac{\frac{[Q] \quad \neg(P \rightarrow Q)}{P \rightarrow Q} (\rightarrow \text{Intro}) \quad \neg(P \rightarrow Q)}{\neg Q} (\neg \text{Intro})$$

**Explanation.** To prove a negative such as  $\neg Q$  we can use proof by contradiction. First assume  $Q$ , then by using  $(\rightarrow \text{Intro})$  we can introduce  $P \rightarrow Q$ . On the other hand, we already have the premise  $\neg(P \rightarrow Q)$ . Given we have reached both  $\phi$  and  $\neg\phi$  we conclude in fact  $\neg Q$ , i.e. our original assumption  $Q$  must be false. **Note.** here we introduce a “ $\neg$ ” hence  $(\neg \text{Intro})$ , which contrasts with the next example of  $(\neg \text{Elim})$  to remove the “ $\neg$ ”.

**Example 6.5**  $\neg P \rightarrow Q, \neg Q \vdash P$ . **Proof.**

$$\frac{\frac{[\neg P] \quad \neg P \rightarrow Q}{Q} (\rightarrow \text{Elim}) \quad \neg Q}{P} (\neg \text{Elim})$$

**Explanation.** Essentially this is just another proof by contradiction. We assume  $\neg P$  and use the premise  $\neg P \rightarrow Q$  which by ( $\rightarrow$  Elim) gives  $Q$ , which together with  $\neg Q$  forms a contradiction, so we may invoke ( $\neg$  Elim) to discharge  $\neg P$ , and conclude  $P$ . Here, instead of introducing a negation as in Example 6.4, ( $\neg$  Elim) discharges the negated assumption  $\neg P$  and concludes the corresponding unnegated sentence  $P$ .

**Example 6.6**  $\vdash P \vee \neg P$ . **Proof.**

$$\frac{\frac{\frac{[P]}{P \vee \neg P} (\vee \text{ Intro}_1) \quad [\neg(P \vee \neg P)]}{\neg P} (\neg \text{ Intro})}{\frac{\frac{\neg P}{P \vee \neg P} (\vee \text{ Intro}_2) \quad [\neg(P \vee \neg P)]}{P \vee \neg P} (\neg \text{ Elim})}$$

**Explanation.** From the empty set  $\emptyset$  we assume  $P$ , for we have no premises to use, and introduce  $P \vee \neg P$ . Next, assume  $\neg(P \vee \neg P)$  to reach a contradiction, leading to the discharge of our first assumption  $P$ , i.e.  $[P]$ , and conclude the interim step  $\neg P$ . We then introduce  $P \vee \neg P$  and again assume  $\neg(P \vee \neg P)$ , another contradiction through which negation elimination discharges the two  $\neg(P \vee \neg P)$  terms to conclude  $P \vee \neg P$ , as required.

TODO - continue to type up my book notes when I get time...